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Cell migration in microengineered tumor environments

Eujin Um,‡[a,b] Jung Min Oh,‡[a,b], Steve Granick[a,c], and Yoon-Kyoung Cho*[a,b]

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Recent advances in microengineered cell migration platforms are discussed critically with focus on how cell migration is influenced by engineered tumor microenvironments, the medical relevance being to understand how tumor microenvironments may promote or suppress the progression of cancer. We first introduce key findings in cancer cell migration under the influence of physical environment, which is systematically controlled by microengineering technology, followed by multi-cues of physico-chemical factors, which represent the complexity of tumor environment. Recognizing that cancer cells constantly communicate not only with each other but also with tumor-associated cells such as vascular, fibroblast, and immune cells, and also with non-cellular components, it follows that cell motility in tumor microenvironments, especially metastasis via the invasion of cancer cells into the extracellular matrix and other tissues, is closely related to the malignancy of cancer-related mortality. Medical relevance of forefront research realized in microfabricated devices, such as single cell sorting based on the analysis of cell migration behavior, may assist personalized theragnostics based on cell migration phenotype. Furthermore, we urge development of theory and numerical understanding of single or collective cell migration in microengineered platforms to gain new insights in cancer metastasis and in therapeutic strategies.

Introduction

A challenge in modern diagnostics is to build constructively on recent findings in cancer biology, showing the critical relations between tumor microenvironment and cancer invasion.[1-3] Surrounding the tumor cells, there exist extracellular matrix (ECM)[4] and flows of blood or interstitial and lymphatic fluids,[5-7] which supply physical force and guidance to the moving cells, as well as extracellular chemokine signals and hypoxia.[8] Therefore for those who aim for the utmost medical relevance, the key scientific problems cannot stop with understanding cell migration behaviour.

There are also various neighboring tumor-associated cells that can promote cancer development, such as vascular cells, fibroblasts, immune cells, and mesenchymal stem cells, which show some changes in their motility and functions in the presence of cancer cells.[9] Vascular cells are required for angiogenesis to supply oxygen and nutrients to the tumor cells.[10] Cancer-associated fibroblasts (CAFs) are considered to modulate tumor progression through cell–cell contacts[11] and secretion of ECM components, growth factors, and chemokines.[12-14] Although much research is devoted presently to understand the crosstalk between immune cells and cancer cells for the purpose of discovering possible therapeutic vaccine,[15, 16] major questions of larger perspective remain unanswered. Not enough is understood about the conditions when immune cells support or prohibit cancer growth.[17, 18] This review provides a critical discussion of methods to investigate quantitatively how this depends on the cell types, activation status and the environmental conditions.

The migration ability of cancer cell is a major trait of metastasis, which is a stage of cancer where cells spread to other parts of the body, which we refer to as invasion.[19] In each step of the metastatic sequence, cell mobility is critical for (i) dissemination from the primary cancer cells, (ii) invasion through the nearby stromal cells and the extracellular matrix, (iii) intravasation into adjacent blood or lymphatic vessels, and (iv) circulation, followed by (v) extravasation into certain tissues or organs. The epithelial–mesenchymal transition (EMT), by which immotile epithelial cells convert to motile mesenchymal-like cells, can be relevant to tumor invasion and the metastatic process.[20-22] Therefore, assessing the migration of cancer cells, as well as other cell types in the neighborhood, can be a critical measure in a comprehensive study of the effects of the tumor environment. (**Fig. 1**)

[a] Center for Soft and Living Matter, Institute for Basic Science (IBS), Ulsan 44919, Republic of Korea
[b] Department of Biomedical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan 44919, Republic of Korea
[c] Department of Chemistry, Department of Chemistry, UNIST, Ulsan 44919, Republic of Korea
* E-mail: ykcho@unist.ac.kr
‡ These authors contributed equally to this work
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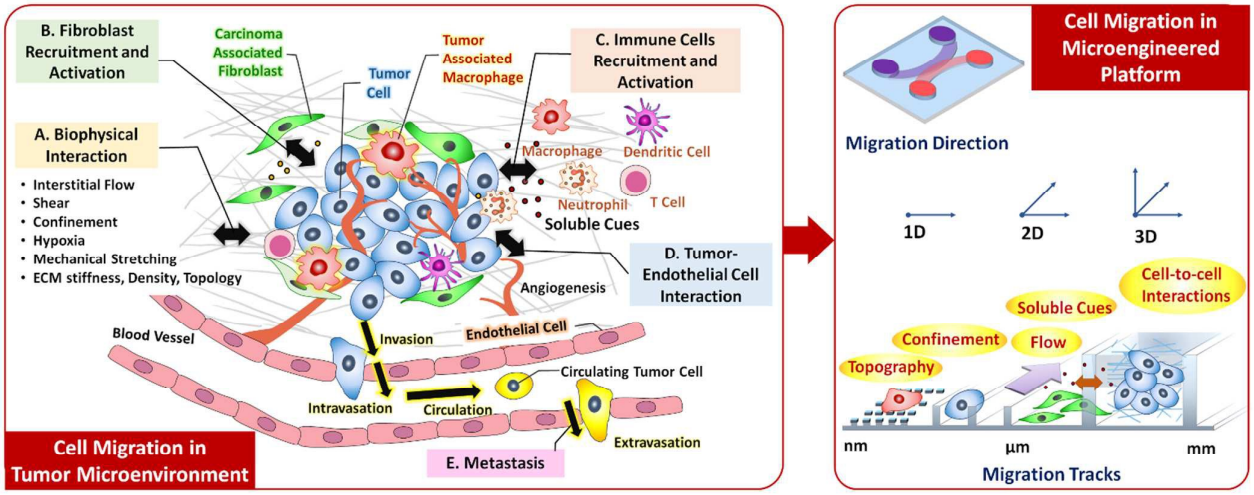


Figure 1. Tumor microenvironments to be realized in microengineered cell migration platform. Within tumor environments the relevant cues include physico-chemical surrounding components, mechanical components such as flow, confinement, topology, and stretching, and chemical components including protein and other signals (A). Other relevant cues include neighboring cells, which migrate towards cancer cells, away from them, or induce cancer cell motility changes (B-D). Other relevant cues include recruitment and activation of fibroblast (B), and immune cells (C), and tumor-endothelial cell interaction that can lead to angiogenesis (D). All these cues can affect the development of metastasis, through which cancer cells obtain the ability to invade the circulatory system and other secondary sites leading to spread of cancer. Schematic illustration on the right shows the microengineered device mimicking the tumor environment, in which the cell migration tracks can guide 1D, 2D, or 3D dimensional movements of the cells and control multiple type of physical and biochemical cues, and cell-to-cell interactions.

The traditional approaches to study cell migratory patterns are *in vitro* on 2D flat surfaces of cell culture plates. However, the flat substrates do not represent the real complex tumor environments. Although *in vivo* methods provide highly realistic data, the findings from such studies demand costly high-tech imaging systems that are unavailable to many,[23] and furthermore the needed monitoring and regulation of several bio-physical/chemical factors simultaneously within the same system have been a formidable challenge.

Microfabrication and microfluidic technology enables the control of variables such as geometries, forces, and material properties with high precision. The most common technology for fabrication of microengineered cell-culture devices is photolithography and replica molding, utilizing photoresist and polymer materials to create geometries in nm to mm scales.[24] With the help of microfluidic technology, the structures can hold various materials either in liquid, solid, or gel forms, and exert forces (in pN to nN) onto the cells cultured in the devices to reflect the complexity of tumor environment **(Fig. 1)**. Therefore, the cell-based assay platforms can be designed to mimic the physical environment of complex tumor surroundings,[25-27] or capture single cell units to closely look at the individual cell behaviours.[28] The microsystems allow efficient analysis of individual cells and their interactions, in a limited space optimized for imaging, while providing biophysical and chemical signals of interest or drugs in accurate quantity for reproducible results. The best current systems either integrate as many as possible multiple known cues in various combinations for cell migration, or study with high control the effects of unknown signals. Such platforms aim for screening the physical and chemical parameters that affect cell migration, in a high-throughput manner.

This review introduces recent developments of microplatforms for cell migration assay, with emphasis on how to establish relevant tumor microenvironments including physical cues, biochemical signal gradients integrated with multiple types of cues, and cell-to-cell interactions. In most of the examples discussed below, cells are embedded within a hydrogel to allow migration inside porous 3D extracellular matrix (ECM) environments, and we note the added advantage that this material is easily integrated into microchannels within which chemoattractant gradients can be established. However, as there are more diverse surroundings in the tumor environment, there are also some channels or tracks for 1D or 2D migration, or surfaces with modified topologies, which will be also be discussed in this review. A final section of this review seeks to anticipate potential uses. We suggest that single cell-based migration platforms can be useful for analysing heterogeneous cancer cells and clinical diagnosis. Furthermore, as further development of theory and simulations of cell migration may help to discern presently-unrecognized patterns of cancer cell migration, this may assist in developing more quantitative and objective means for not only diagnosis but also medical prognosis.

Interaction of migrating tumor cells with physical environments

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While most of the biomedical research has focused on finding direct biochemical signals inside and outside of the cells that affect cancer cell motility, development of microengineering technology helped to broaden the research scope including the effect of physical forces exerted by the surroundings of the cells. Transduction from the external mechanical stimulation to inner biochemical cell signaling that leads to changes in cytoskeletal organization inside the cells is a complex process. Material and physical properties of the ECM, such as stiffness, density, and alignment, have been extensively studied to show their effects on the cell morphology, speed, and persistence of cell migration.[4, 29-31] In the tumor microenvironment, physical factors other than the material properties of ECM itself exist, such as flow, geometrical constrictions, and dynamic changes of the surrounding.[32, 33] Therefore, microfabricated platforms to simplify and control the external forces are advantageous for quantitative analyses of cell migration. In this section, we will particularly focus on the recently reported microengineered devices of μm scale that reveal novel aspects of the tumor cell migration mechanisms with flow and geometrical confinement, and of nm scale with topological stimulation.

Interstitial flow and shear stress

Interstitial flow is the transport of fluid through the tissue extracellular matrix. The pressure gradient induced by interstitial flow established between the original tumor site and normal tissue sites can increase the possibility of tumor metastasis.[7, 34-36] Polacheck *et al.* embedded tumor cells in a collagen gel, and its sides were adjacently connected with microfluidic channels separated by micropillars, mimicking interstitial flow (**Fig. 2A, i**).[35] In this design, the effect from shear force by bulk flow outside the gel matrix could be neglected, so that the authors were able to resolve the directional bias of the cell migration as a function of cell density, CCR7 (chemoattractant) activity, and interstitial flow velocity. The mechanism of cells moving downstream of the flow by CCR7-dependent chemotaxis competes with flow-induced tension-mediated upstream movements.

Direct flows over cells, such as blood stream, lymphatic flow, and interstitial flow, will induce a shear stress on the cells, which also plays an important role in the cell migration and cellular signaling related to tumor proliferation and metastasis.[37, 38] Rizvi *et al.* showed the growth of ovarian cancer nodules under continuous laminar flow in a microfluidic channel, and observed that shear stress induces cancer cells to have motile and aggressive tumor phenotypes.[6] Increased expression of epidermal growth factor receptor (EGFR) and vimentin and decreased expression of β-catenin protein indicate epithelial-mesenchymal transition and metastatic potential (**Fig. 2A, ii**).

Geometrical confinement

Hydrogel materials embedded with cells have been extensively used in microfluidic cell platform to create 3D spaces for cell migration studies, apart from conventional 2D open substrates such as glass and Petri dishes.[12, 39] However, fibers, bundles, and tunnels of ECM or muscles, vascular microtracks, and interstitial spaces of epithelial or endothelial tissues provide more complex geometrical confinements, even in 1D and 2D spaces as well as 3D spaces. Confinement, in a broad sense, means giving restrictions in the cell movements either by geometrical constriction or changing the adhesive forces. Microfabricated grooves on substrates resulted in clear directional preferences of the cells,[40] and contact confinement in various geometries via surface protein patterning affected

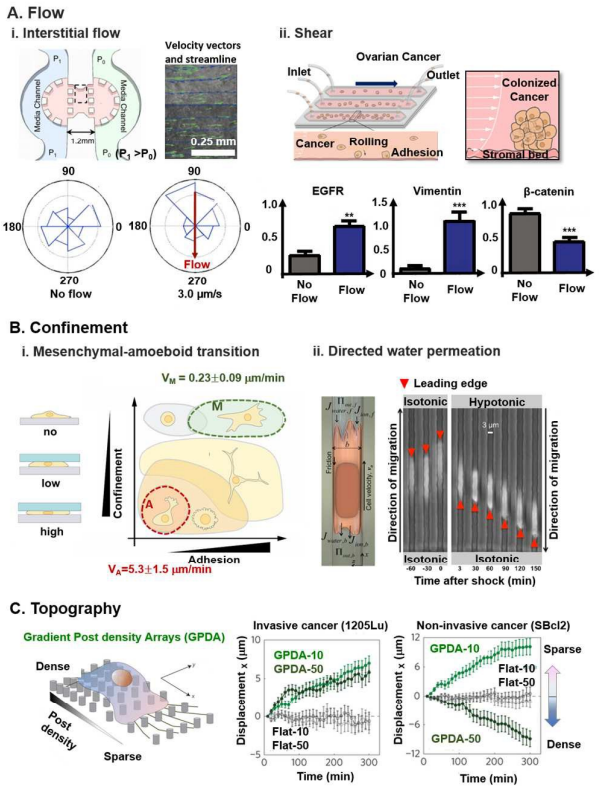


Figure 2. Migrating tumor cells interacting with the physical environment. **A. Flow effect:** (i) Interstitial flow in a matrigel was mimicked by two side channels with a pressure difference, which induced a directional bias in cellular migration. The polar histograms compares the random migration of the cells in control (left) and the migration against the flow direction (right). (reproduced from ref. 35 with permission from the National Academy of Science). (ii) Shear stress increased metastatic characteristics of ovarian cancer cells cultured in 3D, in phenotype and in post-transcriptional expression, affecting the migration (reproduced from ref. 6 with permission from National Academy of Science)**. B. Confinement effect:** (i) Slow mesenchymal cells can switch from phenotype A (normally spread cell) to M (round cell with a leading edge), exhibiting fast amoeboid-like migration under conditions of low adhesion and strong confinement (reproduced from ref. 48 with permission from Elsevier). (ii) The osmotic engine model demonstrates cancer cell migration by water permeation in a 1D confined microenvironment. Differences in osmotic pressure between the front and back side of a 1D confinement channel can elucidate the role of aquaporins and ion channels in cancer migration (reproduced from ref. 50 with permission from Elsevier). **C. Topography:** Topotactic migration of invasive melanoma cells (1205Lu), guided by the gradient of the post density arrays (GDPA) showed migration directionality towards sparser post array zones, whereas non-invasive melanoma cell lines (SBcl) showed fibronectin-density dependent migratory behaviors. The number 10 and 50 next to GPDA and Flat in the figure legend indicate the concentrations of coated fibronectin in μg mL^{-1} (reproduced from ref. 52 with permission from Nature Publishing Group).

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migration-related molecules, with increased metastatic characteristics and tumorigenicity.[41] The presence of ceilings or different wall angles in the cell migration track also revealed new mechanisms of cell migration.[42] Therefore, confinement in cellular level is revealed to be an actual physical cue that modulates intracellular signaling in cell migration, which is an environment to be mimicked by microscale engineering.[43, 44]

Cancer cells even degrade the surrounding ECM, and create tracks for their migration. The plasticity of cancer cells not only remodels the surrounding tissues, but also change the innate cellular migration behaviors from collective to individual forms, or exhibit epithelial to mesenchymal and mesenchymal to amoeboid transitions, under the influence of signals from microenvironment.[22, 45, 46] The degree of confinements surrounding the cells is also found to influence the change in the migration modes of cells.[47] Liu *et al.* made a schematic diagram of the cell phenotypes as a function of the confinement height and adhesion strength (**Figure 2B, i**).[48] Confinement channels usually refer to 2D substrate with a low ceiling, or 1D track limiting the cell movement to a single axis. The dimensions of height or width of the channels are typically in the range of 3 to 10 μm. Among them, normally spread cells (M) can switch to fast (about 8 times faster) round cells with a leading edge (A) under conditions of low adhesion and strong confinement by the process called mesenchymal-amoeboid transition (MAT). Even in the absence of external chemotactic gradients or the presence of the drugs targeting cytoskeletons for reduction in the migration speed, the confinement spontaneously induced persistent motility in some cell population.[49] This mechanism proposes an adaptation of cancer cells to escape primary tumors and invade tissues without any specific genetic alteration.

One of the novel findings of cell migration from 1D confinement track is osmotic engine model, describing the cell movement by cell-volume regulation via water permeation (**Fig. 2B, ii**).[50] Cell lines from both of carcinoma and sarcoma confined in a narrow channel, establish a polarized distribution of active ion pumps and aquaporins in the cell membrane, which creates a net inflow and outflow of water and ions at the cell's leading or trailing edge leading to the net cell displacement. Actin polymerization may help establish initial ion-channel and aquaporin polarization, but the major mechanism enabling the migration is water permeation in confinement rather than other cytoskeleton dynamics. This microengineered platform helped find one of the important mechanism of how cancer cells migrate in narrow constrictions of 1D tracks *in vivo*, and this platform can also be utilized in finding cancer therapeutic treatment, targeting aquaporins and ion pumps. Recently, researchers have started to integrate the confinement design in chemotaxis experiments, which will be introduced in the next section regarding multi-cue environments.

Topography

Topotaxis is the directional migration of cells guided by the gradient of nanoscale textures on the substrate. The term can be differentiated from haptotactic- or durotactic-directed (rigidity) cell guidance.[51] Experiments on the migration of melanoma cells guided by the gradient of post density and pre-coated ECM density was performed to propose that the migration capacity of many cancers may depend on topotactic responses (**Fig. 2C**).[52] The invasive melanoma cell lines examined (1205Lu and WM983B) showed migration towards sparser post array zones independent of the fibronectin density, whereas the direction of two non-invasive melanoma cell lines, (SBcl2 and WM1552) was fibronectin-density dependent. These results suggest that while cell migration following the gradient of topology is greatly influenced by the material properties of cells and the surrounding ECM, the melanoma cells are more sensitive to the density gradient of topographic features.

Recent examples of on-chip models of chemotaxis: multiple migratory cues in the tumor environment

Directional migration is induced by several cues *in vivo*: chemotaxis, durotaxis, haptotaxis, galvanotaxis or simple contact guidance.[53] Investigation of one type of migration cue is good for quantitative measurements and simplifying the model, but it does not provide enough information about competitive or synergetic effects of multiple-cues simultaneously present. In this section, we will particularly focus on the recent development of microengineered platform to investigate multi-cue environments that are close to the real tumor migration system.

Many conventional research platforms for chemotaxis are built based on the assumption that cells sense extracellular cues of interest, such as cytokines, hormones, or microvesicles, in soluble form in the liquid medium.[54-57] The basic tool to test directed migration is a Boyden chamber, which separates migrating cells across a porous membrane towards the chemokine, away from non-migrating cells. However, in this kind of a source and sink design, the gradients fluctuate and change over time which hinders quantitative measurements of the effective concentrations, and observation of cell trajectories through a membrane is nearly impossible.[58] Using a microfluidic system, a continuous flow of the chemokine can be established, which helps establish either a constant or a dynamic concentration gradient, or gel can be integrated for more stable maintenance of the gradient.[56]

Surface-bound protein gradients are also omnipresent *in vivo* on the ECM or on tissue surfaces. Haptotaxis is the term to describe the directionality of cells following solid-bound chemokines, and the combination of chemo- and haptotactic guidance cues play fundamental roles during various physiological processes such as protective immunity, tissue regeneration, and pathogenic cancer metastasis and autoimmunity.[51, 59] Gradients of surface-bound proteins can be realized in *in vitro* experiments with precisely controllable surface-functionalization techniques. An example of incorporating both chemotaxis and haptotaxis is shown in **Fig. 3A.**[60] In this paper, one of the immune cells, dendritic cells, are put in a chamber where surface-bound CCL21 proteins form a gradient using laser-assisted adsorption by

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photobleaching, and an opposing gradient of CCL19 in solution is also introduced by a microfluidic valving source-sink system. Depending on the relative strength of each gradient, the direction of cell migration changes as indicated in the Rose plot of angle distributions of all tracks. This technique can be applied in tumor migration representing a more *in vivo*-like environment where solid-state chemokines such as cell surface or connective tissue-bound growth factors exist together with soluble chemokines.

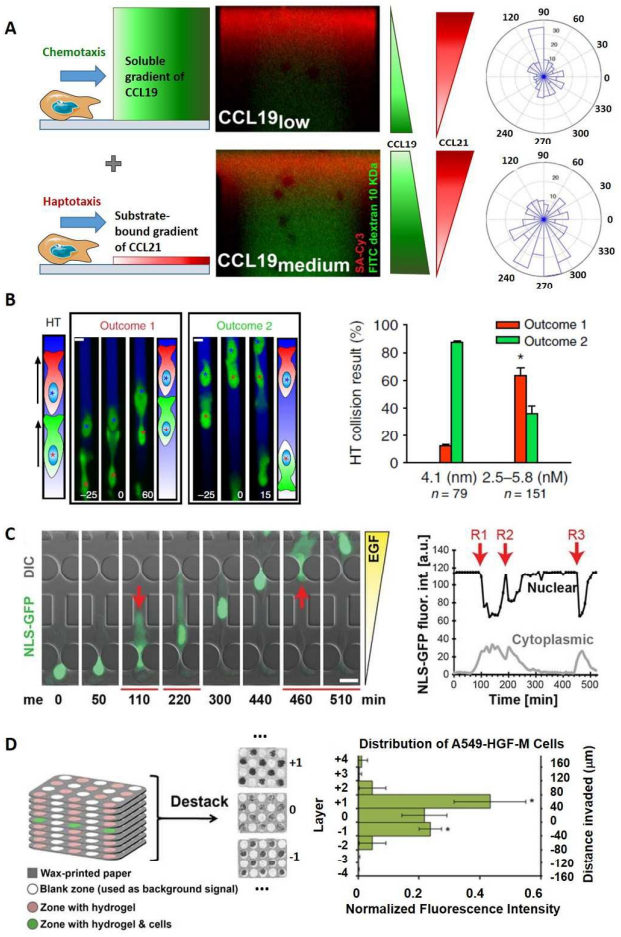


Figure 3. Chemotaxis with multiple migratory cues. A. Combination of substrate-bound and soluble chemokine gradients: Dendritic cells migrating on opposing chemotactic CCL19 and haptotactic CCL21 gradients show different directionality depending on the strength of each attractant. The combination of chemo- and haptotactic guidance cues may reveal more realistic cell migration behaviors during cancer metastasis under competitive multi-signals (reproduced from ref. 60 with permission from Nature Publishing Group). **B. Interplay between chemotaxis and contact inhibition of locomotion:** Head-to-tail collisions in breast cancer (MTLn3-B1) cells in 1D channels under gradients of EGF, resulting in Outcome 1 (no changes) or Outcome 2 (contact inhibition resulting in switched polarity of the cell) (reproduced from ref. 61 with permission from Nature Publishing Group). **C. Chemotaxis in tight spaces:** Representative image sequence of breast cancer cells (MDA-MB-231) experiencing extensive deformation with multiple nuclear envelope ruptures and restoration while moving through narrow constrictions (reproduced from ref. 62 with permission from The American Association for the Advancement of Science). **D. Chemotaxis in a paper-based invasion assay:** a paper and hydrogel-based method that can isolate sub-populations of cells based on their invasiveness in a gradient of concentration of oxygen (O_2) along with other possible candidates in the cell media (reproduced from ref. 63 with permission from Elsevier).

In the migratory path, cells may encounter physical obstacles, including neighboring cells. The second example demonstrates the effect of contact with other cells during chemotactic migration (**Fig. 3B**).[61] Cells will encounter other moving or stationary cells during migration which may hinder their persistence, which the authors call "contact inhibition of locomotion (CIL)." To simplify, they created 1D channel tracks where cells can collide head-to-tail or head-to-head in gradients of epidermal growth factor (EGF). Cells simultaneously process chemotaxis and CIL cues, and the example of a head-to-tail collision (**Fig. 3B**) results in two outcomes where CIL does not occur (Outcome 1) or CIL changes the direction of the following cell (Outcome 2). The probabilities of the two outcomes change depending on the strength of the concentration gradient. They found that this contact is not only a physical barrier but also that interactions with the neighboring cells take place. For example, when fibroblast (3T3) cells were in the way of cancer cells (MTLn3-B1 cells), the majority of chemotactic cancer cells migrated around and actively displaced the fibroblast cells, instead of changing direction. This dynamic crosstalk may promote tumour invasiveness by preferentially directing cancer cell migration into the stromal environment, or in clusters.

As mentioned in previous sections, a degree of confinement similar to the cell size changes the cell migratory behavior. Denais *et al.* constructed a platform in which cells encounter confinements less than their nucleus size (5 × 2 μm^2) during chemotactic migration (**Fig. 3C**).[62] Surprisingly, nuclear envelopes rupture and repair during the migration as shown by the fluorescence intensity of loss and gain by the nuclear and cytoplasmic nuclear localization sequence (NLS-GFP) of the cells. In fact, during cancer metastasis, tumor cells penetrate tissues through tight interstitial spaces, requiring extensive deformation of the cell and its nucleus.

The invasiveness of the cancer cell can be determined by the distance migrated through stacks of hydrogel-infiltrated paper (**Fig. 3D**).[63] The unique feature of this assay is the ability to isolate cells by the distance of migration and easy access of the chemical cues into the 3D stacks from both the top and bottom, to form multi-cue gradients from either side. The effect of oxygen gradient, which is related to hypoxic conditions found in invasive tumor environment,[64, 65] was comprehensively investigated along with other nutrient gradients in the medium i.e., glucose, autocrine factors, and waste products). The study of multi-cue microenvironments with the appropriate design of a microfluidic platform takes us one step further in comprehensively understanding actual cell migrations *in vivo*.

Intercellular migration in the tumor microenvironment

Cell-to-cell communication in the tumor microenvironment is primarily mediated by various paracrine factors across the

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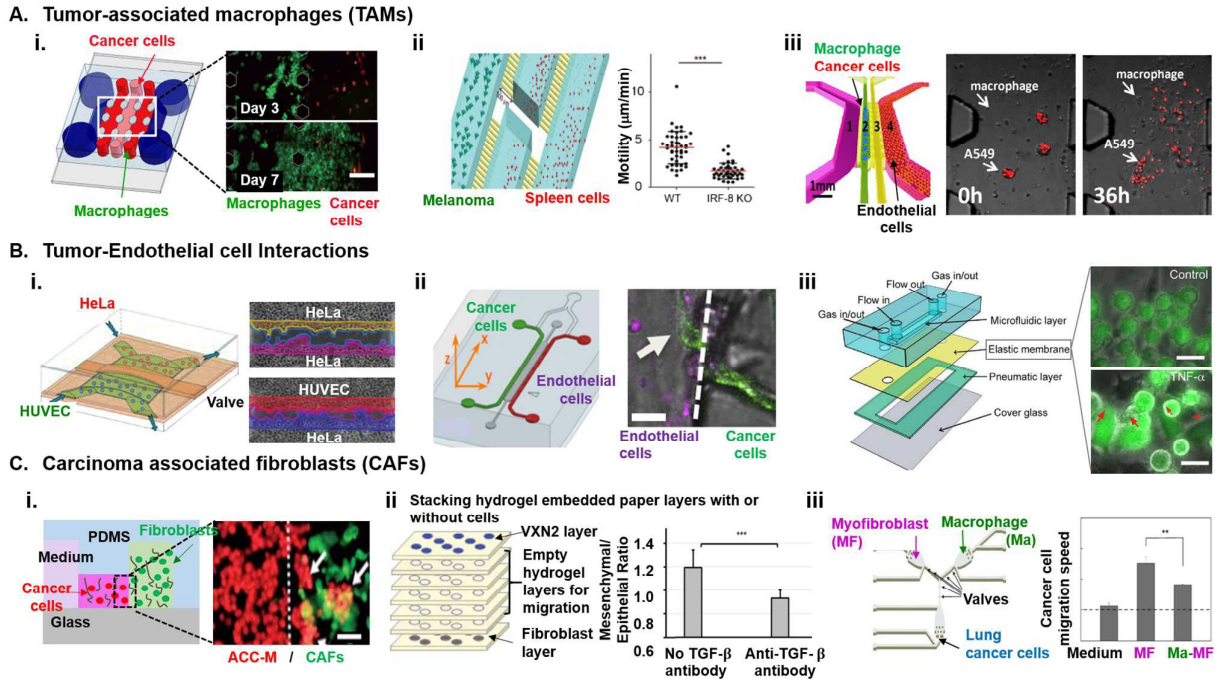


Figure 4. Intercellular migration through cell-to-cell communications. A. Tumor-associated macrophages (TAMs) (i) GFP-labeled macrophages (RAW 264.1) cultured in a matrigel invaded into neighboring collagen gels containing breast cancer cells (MDA-MB-231) but not into gels without cells, over a 7-day culture experiment. Scale bar: 200 μm (reproduced from ref. 75 with permission from the Royal Society of Chemistry). (ii) When splenocytes (heterogeneous populations of primary immune cells) and melanoma cells (murine B16-F10) were placed on each side of a microfluidic device, splenocytes migrated toward melanoma cells. When IRF-8 KO immune cells with less motility were used, B16 cells showed a more invasive behavior (reproduced from ref. 71 with permission from the Royal Society of Chemistry). (iii) In a 3D microfluidic platform with tumor cell aggregates (A594) co-cultured with macrophages and HUVECs, macrophages showed preferential migration toward tumor aggregates, which were more dispersed in contact-based communication with macrophages after 36 hours (Scale bar: 100 μm) (reproduced from ref. 76 with permission from Impact Journals). **B. Tumor-endothelial cells** (i) Images of co-culture chambers with HeLa to HeLa cells or HeLa cells to HUVECs, obtained 12 h after the co-culture (reproduced from ref. 80 with permission from the American Chemical Society) (ii) Intravasation of breast carcinoma cells (MDA-MB-231, green), indicated by white arrow, through the endothelium (HUVEC, magenta), induced by TNF-α-secreting macrophages (RAW264.7, not shown in figure). Scale bar: 30 μm (reproduced from ref. 81 with permission from the National Academy of Science) (iii) Invasive protrusion of cancer cells (HeLa, green) on HUVECs in TNF-α conditioned medium (Scale bar: 20 μm) (reproduced from ref. 77 with permission from Nature Publishing Group). **C. Communication with fibroblast cells** (i) In a hydrogel-based 3D co-culture platform, MMP-secreting carcinoma-associated fibroblasts (CAFs) (green) promoted the invasion (indicated with white arrows) of salivary gland adenoid cystic carcinoma (ACC) cell spheroids (red) (Scale bar: 100 μm) (reproduced from ref. 89 with permission from the Royal Society of Chemistry). (ii) Fibroblasts enhanced migration of human lung cancer (VXN2s) cells within the stacks of 100 μm thick-paper scaffolds by TGF-β secretion with increased EMT expression (high ratio of vimentin to E-Cadherin). Blocking TGF-β decreased EMT expression of the VXN2 cells (reproduced from ref. 90 with permission from John Wiley and Sons). (iii) A microfluidic device with microvalves to regulate the inflow of paracrine factors from three different cell types (lung adenocarcinoma CL1-0, lung fibroblast MRC-5, THP-1 monocyte derived M2 macrophage) in which the enhanced cancer migration speed (normalized) in the presence of myofibroblasts or M2 macrophages was observed (reproduced from ref. 91 with permission from the Royal Society of Chemistry).

media or by direct contact between cells.[57, 66-68] Various multiple factors from tumor cells recruit many tumor-infiltrating cells, including tumor-associated macrophages (TAMs) and neutrophils (TANs), lymphocytes, endothelial cells, cancer-associated fibroblasts (CAFs), and mesenchymal stem cells (MSCs).[9, 57] Tumor-infiltrating cells can support cancer progression by inducing angiogenesis, immune invasion, and metastatic dissemination, creating complex migratory tracks in tumor microenvironments.[69]

To study cell-to-cell interactions in tumor microenvironments, microfluidic chips can provide a platform to place neighboring cells of single or multiple types adjacent to each other, controlling the distances between them. Quantitative and controlled cell-to-cell environments will help to understand the complexity and heterogeneity of cell migratory behaviors. In this section, we will introduce recent examples of microplatforms for investigating cell migration in the presence of macrophages, endothelial cells, or fibroblasts, which co-function to represent the complexity of tumor environments.

Macrophages

Immune cells, such as macrophages and dendritic cells, tend to infiltrate the tumor microenvironment, and act to suppress or promote cancer progression, though the mechanisms are not clearly understood.[70, 71] For example, mouse spleen cells migrate and interact with dying fibrosarcoma cells with an annexin-1 ligand, via formyl peptide receptor 1 (FPR1), thereby enhancing anti-tumor immune responses.[72] However, once tumors overcome the threshold of the immune barrier, they begin to modulate immune cells in a way to support tumor development.[3] Tumor-associated macrophages (TAMs) can constitute up to 50% of the tumor mass,

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and they can promote tumor metastasis by suppressing the cytotoxic functions of natural killer (NK) cells and CD8+ T cells and secreting interleukin-4 (IL-4) and IL-13 to help tumors progress.[73, 74] Therefore, communications between those cells need to be intensively analyzed to correctly understand their relationships.

The microengineered platforms for studying communications between immune cells and cancer cells are designed for observing the motility patterns in immune cell migration toward cancer cells and study the effect of critical factors in regulating tumor development. When MDA-MB-231 metastatic breast cancer cells and RAW 264.1 macrophage cells were patterned adjacent to each other inside a microchannel (**Fig. 4A, i**), the co-cultured macrophages exhibited a greater tendency for migration toward breast cancer cells during 7 days of culture, compared with the mono-cultured macrophages.[75] The proliferation rate of macrophages was also enhanced by cancer-derived paracrine factors. Heterogeneous immune cell populations, which were cultured apart from murine B16-F10 melanoma cells with a middle microchannel in between, migrated through the channels towards B16 cells and formed clusters of activation (**Fig. 4A, ii**).[71] When the immune cells from IRF-8 KO (interferon regulatory factor 8 knock out) mouse were integrated, they poorly interacted with the melanoma cells. On the other hand, B16 cells become more attracted towards the microchannels. Bai *et al.* combined three cell types, cancer, endothelial (HUVEC monolayer), and immune cells in microfluidic channels (**Fig. 4A, iii**), where different subsets of macrophage showed abilities to migrate towards human lung adenocarcinoma (A594) aggregates and induce dispersion of the aggregates via contact-based interaction between ICAM-1 and β2 integrin.[76] The expression levels of E-cadherin in the cancer cell aggregates were also decreased indicating the promotion of EMT. Similar results were observed when HUVECs were excluded from the microchannel, suggesting that the endothelial cells do not alter the response of these subtypes of macrophages.

Endothelial cells

Tumor-endothelial cell interactions are important in multiple steps of cancer progression, such as angiogenesis and metastasis.[77-79] Microfluidic co-culture platforms have been employed to investigate the complex cellular interactions. For example, Zheng *et al.* placed HeLa cells and human umbilical vein endothelial cells (HUVECs) in adjacent chambers which were separated by a 300 μm distance and a pneumatic valve (**Fig. 4B, i**).[80] The microvalve helps the starting position and time of the cell migration to be constant from experiment to experiment. Upon opening of microvalve to connect the chambers, migration of the two cell types toward each other was observed. The migration rate of tumor cells cultured in the presence of HUVECs was enhanced compared with the result of mono-cultured tumor cells. They also observed the retreat of HUVECs upon the approach of HeLa cells during co-culture, indicating that soluble factors secreted from HeLa cells play a critical role in repelling the endothelial cells in the co-culture system.

Zervantonakis *et al.* demonstrated a 3D microfluidic chip to model the tumor-vascular interface. Two independent microchannels, one seeded with endothelial (HUVEC) cells and the other with breast carcinoma cells (MDA-MB-231), were interconnected via a 3D ECM hydrogel, where intravasation of the cancer cells through the ECM matrix was investigated (**Fig. 4B, ii**).[81] Although intravasation was a rare event that occurred for a small fraction of the tumor cells in contact with the endothelial monolayer, it was significantly increased by introducing macrophages, which secrete tumor necrosis factor alpha (TNF-α) when in contact with the endothelial cells, resulting in endothelial barrier impairment.

Huang *et al.* designed a microfluidics-based tumor extravasation model, integrated with cyclic airflows to investigate synergistic effects of mechanical and biochemical stimulation (**Fig. 4B, iii**).[82] Blood mixed with a small amount of HeLa cells was introduced into endothelial cells (ECs)-pre-coated microchannel, and they created a hemodynamic background by inducing fluid shear stress (FSS) and cyclic stretch in the device. When the intensity of shear stress and cyclic stretch was mimicked in the capillary with the condition similar to a major or medium-sized artery, adherence of HeLa cells on the EC layer was promoted. Furthermore, the integrity of the EC monolayer was destroyed by TNF-α conditioned medium, which facilitated tumor cell adhesion and infiltration. The device also tested the effect of a drug candidate for reducing ROS (reactive oxygen species) by the administration of platinum nanoparticles (Pt-NPs), and the integrity of the ECs monolayer was protected to prevent the invasion of tumor cells.

Fibroblasts

Cancer-associated fibroblasts (CAFs) are reported to promote cancer survival and growth, enhancing the rate of cancer invasion.[83] CAFs were demonstrated to support angiogenesis in tumor, increase drug resistance, and suppress the immune responses of cancer cells. The soluble factors secreted from fibroblasts, such as transforming growth factor β1 (TGF-β1), insulin-like growth factor (IGF), hepatocyte growth factor (HGF), platelet-derived growth factors (PDGFs), interleukin-6 (IL-6), and matrix metalloproteinases (MMPs), may be indirectly involved in promoting cancer invasion. They are also known to alter production, digestion, and organization of the ECM, in a direction to help cancer invasion and metastasis.[13, 84-86] To investigate the effect of ECM reorganization on cancer cell migration, Ao *et al.* used a vacuum chamber-assisted microdevice to mechanically stretch fibroblasts, which modified their alignment of fibronectin structures from a random to a more linear direction.[87] They examined the ability of stretched fibroblasts to direct migration of neighbouring cancer cells (SCC61) in a more persistent way by co-culturing them.

Furthermore, invasive migration of the colon cancer cells through the collagen gels was shown only when CAFs were added.[88] The proinvasive activity was found also with conditioned media from CAF cultures, suggesting the involvement of soluble mediators in CAF-promoted cancer invasion. Tumor spheroid invasion induced by CAFs has rarely been observed *in vitro*, and Liu *et al.* tested the

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spheroidal forms of salivary gland adenoid cystic carcinoma (ACC-M) cells by embedding them in a matrix inside a microchannel.[89] The spheroids were shown to invade the CAF-containing region of the matrix, but not the regions with normal fibroblasts or empty areas (**Fig. 4C, i**). Camci-Unal *et al.* co-cultured primary human lung tumor cell lines (VXN2s) with fibroblasts in stacks of matrigel-embedded paper scaffolds (**Fig. 4C, ii**).[90] The number of migrated cancer cells within the paper showed that TGF-β secreted from the fibroblasts enhanced the migration of the tumor cells, and the inhibition of TGF-β decreased EMT for the tumor cells.

The effects of TGF-β secreted by cancer cells, as well as the influence of another cell type, such as immune cells, were also tested on the migration of cancer cells. A chip with active pneumatic microvalves was designed to regulate the diffusion of paracrine factors secreted from three different cell lines: human lung adenocarcinoma cells (CL1-0), lung fibroblast cells (MRC-5), and M2 macrophages differentiated from a monocyte cell line (THP-1) (**Fig. 4C, iii**).[91] TGF-β1 secreted from the cancer cells induced the differentiation of fibroblasts to myofibroblasts, and cytokines secreted from myofibroblasts enhanced the migration speed of the cancer cells. Soluble factor derived from macrophages also had an enhancing effect on the speed of cancer cells. However, TNF-α produced by macrophages reduced the expression of α-smooth muscle actin and the secretion of TGF-β1 in myofibroblasts, which in turn, reduced effects of myofibroblasts on the speed of cancer cell migration.

The microfluidic platform for cell-to-cell communications suggest that synergistic or competitive effects of cytokines from multiple cell types definitely alters the migration behaviors of the cells in tumor environments. Since the migrating entity could be either cancer cells, or other immune/stromal cells, creating an optimal environment for the growth of each specific cell type and controlling the position, while integrating multiple cell types is an important advantage of using the microengineered platform. Although current microengineered platforms enabled visualization of the migration of cancer cells or other cell types upon their interactions, most of the studies still do not clarify cancer derived factors responsible for stimulating the intercellular communication. Therefore, the platforms for immune based anti-tumor therapeutics and drug screening are at very early stage of development,[92, 93] and microfluidic platform design needs to reveal more in-depth dynamics of the cell migratory patterns and their sophisticated communication with cancer cells.

Beyond multi-cue tumor microenvironments and future perspectives

In previous sections, we introduced some of the advances in microplatforms used to investigate the intricate movement of cells in tumor microenvironments with high precision. Now, beyond the step of understanding the phenomenological state of cell migration in a limited experimental setup, more physically and physiologically relevant systems will emerge in the future.

The microengineering system has proven to be very effective for single-cell isolation and analysis, although the throughput in analysis can be improved.[28, 94] Even though there have been remarkable progress in microengineering platforms in creating multi-cue microenvironments for

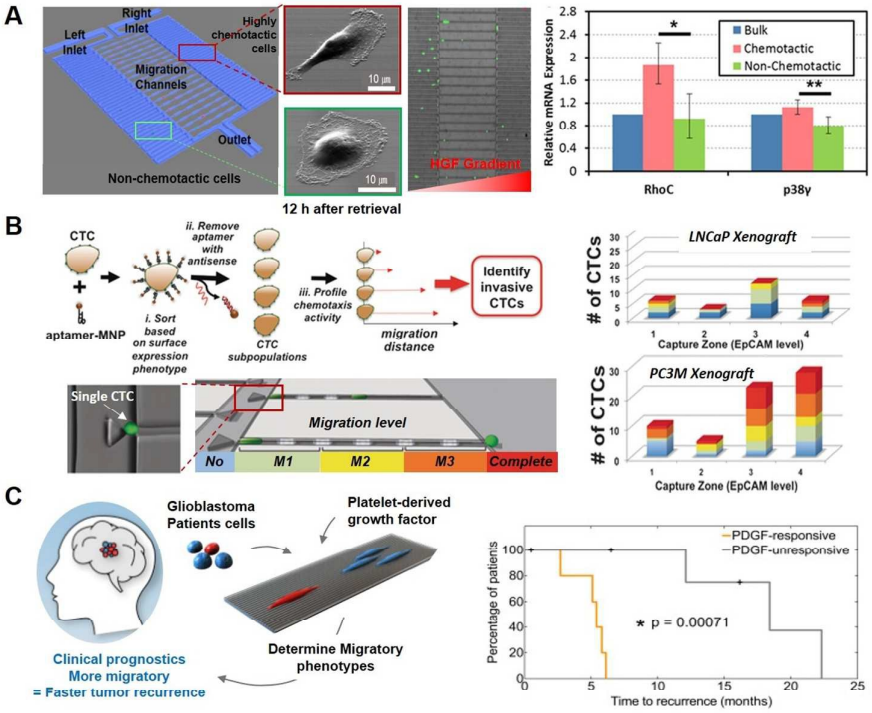


Figure 5. Sorting of heterogeneous cancer cells after migration. A. A microchannel for single-cell based migration within hepatocyte growth factor (HGF) concentration gradients and isolation enabled separation of highly chemotactic ovarian cancer cells (SKOV3) with mesenchymal morphology from non-chemotactic cells exhibiting epithelial morphology. The chemotactic cells also exhibited a higher expression of mRNA for metastasis-associated genes RhoC and p38γ as compared to non-chemotactic cells (reproduced from ref. 97 with permission from Nature Publishing Group). **B.** Circulating tumor cells (CTCs) first isolated according to the expression of a surface marker (EpCAM) were further profiled into subpopulations corresponding to migration distances in response to a chemoattractant (CXCL16). CTCs extracted from mice in cancer xenograft models were categorized as either M1 (minimal migration), M2 (intermediate migration), M3 (high migration), or migrated (reproduced from ref. 98 with permission from John Wiley and Sons). **C.** A glioblastoma multiform migration platform with a fibrillar surface was used to screen 14 patient samples. The migratory phenotype of a subset of cells in response to platelet-derived growth factor was highly predictive of tumor location and recurrence in the clinic (reproduced from ref. 99 with permission from Elsevier).

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studying cell migration, only a few research has emerged combining the single cell sorting technology with distinctive cellular migrations, utilizing cell motility characteristics as theranostic tools. In addition to that, more in-depth understanding of the biophysical principles governing the migration mechanism will be inevitable with the development of the microengineering system. The current theoretical and numerical analysis are far less than understanding the complex and sophisticated results of experiments. Here, we introduce some of the novel microplatforms for translational research of single-cell sorting and analysis based on the migration ability. We further suggest a few examples of the theoretical/numerical modelling of cell migration that can be applied to the analysis of experimental results from the microengineered platforms as future prospects.

Clinical analysis of the cells sorted by distinctive migratory behaviors

Cancer cells are heterogeneous in phenotype and genotype even when from the same original tumor site.[66, 95] In the analysis of cell migration, the results are mostly based on the ensemble averages of the whole population, in which the specific dynamics of rare cells can be neglected. The accurate and refined analysis of single cells emerging as a major research field, and since the development of fluorescence-activated cell sorting (FACS), more advanced techniques to capture and isolate rare cells without damage are being developed.[96] Still, single cell-based sorting and analysis can be very sophisticated and time-consuming. The study of cell migration in a tumor environment can propose a novel platform to sort cells into groups based on their migratory behaviors. Cell separation can be automatically conducted by cell variances in migration distances, which will be beneficial in cell retrieval without any tagging of the marker. After sorting the cells, detailed chemotactic and chemokinetic phenotype characterizations can be performed.

Very recently, efforts to distinguish cancer cells based on their migration behaviors were successful with microfluidic devices, which show promising results leading to clinical diagnosis and applications. Chen *et al.* combined single cell trapping and migration design in a microfluidic device which could separate migrated cells at the end point based on their chemotactic motility **(Fig. 5A)**.[97] A separate analysis of highly motile cells and less motile cells showed the difference between their morphology and mRNA expression of metastasis-associated genes. Poudineh *et al.* combined two mechanisms of cell sorting: one based on the expression level of a surface marker and another by the migration distance towards a chemoattractant **(Fig. 5B)**.[98] With this microdevice, they could subdivide the cell types and examine them in more detail to compare the characteristics of different tumor xenografts. Clinical analysis with a patient-derived glioblastoma was also conducted by Smith *et al.*, based on migration towards platelet-derived growth factor (PDGF) on nano-scale ridges **(Fig. 5C)**.[99] The responsiveness towards PDGF, measured by the directionality and speed of the cells, correlated with the tumor location and recurrence after surgical resection. The migration-based cell sorting platform can be applied not only to tumor cells but also to other highly motile cells such as immune cells, in which motility can be an important indicator of pathological disease.

For implementing theoretical and numerical analysis

While the experiments on the cell migration can only be performed in limited spatial dimensions and time (typically on the order of a few hundred micrometers in a few hours), theoretical and numerical analyses can promote a more fundamental understanding of the cell migration from a biomechanical point of view.[47, 100] As the microengineered cell-migration system becomes more complex with diverse dimensions and geometry, the physical understanding becomes more important, but currently only a few microengineered platforms employ the theoretical background in their designs. Appropriate modelling of the cell migration may reveal interesting patterns of cell movements in the tumor environment, unrecognized by simple observations. In this section, we want to introduce some of the models and numerical methods that can be applied to the experiments of cell migration on the microplatforms.

The numerical analysis of cell migration can be roughly divided into three categories. One is the stochastic analysis of the observations from single cell migrations. In this case, the study focuses on the motility itself, so statistical measures such as mean squared displacement (MSD), average speed, and directional persistence are compared with various experimental conditions.[101-104] Han *et al.* showed down regulation of tumor cell migration on the nano-roughened surface via MSD and average migration speed by particle in velocimetry (PIV) analysis (**Fig. 6A**).[104] Brownian dynamics based on the Langevin equation is a fundamental tool to interpret the data, in which we assume that the cell predominantly moves in random directions.[105, 106] Through this approach, we can understand if cell migration is diffusive (Brownian), subdiffusive, or superdiffusive according to the microenviroments.[107] The Brownian random motion of tumor cells plays an important role in migrating and searching for targets, but the decision process between diffusive and ballistic movements is still unknown. The searching strategy of a cancer cell can be explained in terms of an intermittent search scheme or a fractional dynamics approach, which are described by non-Markovian statistics or the Fokker-Planck description. Even though this approach is valuable in categorizing the patterns of cell migration, the complex interactions between various attractants and receptors inside and outside cells are not considered.[108-111] Moreover, considering the microenvironment of the migration platforms such as confinement, roughness and adhesion will add more complexity to the modelling, and so far there has not been much consensus on these parameters. In particular, tumor-specific behavior has not been studied extensively.

Another approach to the modelling of cell migration focuses on the morphological dynamics of a single cell during movements. Flexible moving objects on solid substrates can be

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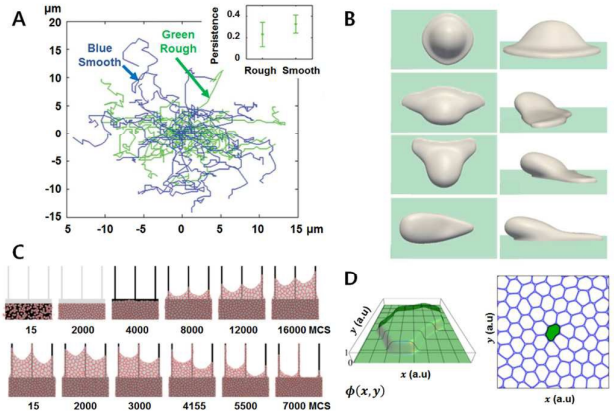


Figure 6. Theoretical and numerical analysis A. PIV analysis of a migrating HuH7 cell layer and individual cell tracking shows that nano-roughness downregulate cell-substrate and cell-cell adhesion proteins in collective tumor cell migration. Blue and green lines indicate smooth and rough surface, respectively (reproduced from ref. 104 with permission from the Royal Society of Chemistry). **B.** A minimal physical model (spreading and retracting by surface tension variations) captures spreading and crawling cell morphologies (reproduced from ref. 115 with permission from Nature Publishing Group). **C.** Generalized Potts models describes the dynamics of collective cell migration on adhesive micropatterns. (Top) Cells cannot migrate to the gray parts of the pattern until the barrier is removed at 4.000 Monte Carlo sweeps (MCS). Cells migrating along the stripes are slowed down by the contractile bridge formed between the stripes. (Bottom) Knockdown of cadherins after 2000 MCS results in a collapse of the bridge. (reproduced from ref. 119 with permission from Albert, Schwarz). **D.** Elastic mismatch between cells enhances motility. Green cell represents a cell with weaker stiffness compared to the surrounding (blue) (reproduced from ref. 114 with permission from Nature Publishing Group).

described by the diffuse interface model without remeshing.[112-116] Tjhung *et al.* showed that a simplified model consisting of an active polar fluid with contractility can successfully represent a cell crawling on a solid substrate (**Fig. 6B**).[115] Here, the directionality of cell crawling is expressed in terms of a polarization field. The local deformations of protrusion and contraction are implemented by changing the local stiffness of a cell membrane and interaction energy between the cell and the substrate. This kind of quantitative analysis of morphological changes of the cell is crucial when studying the interaction between cells and microenvironments. In a very recent study, Ribeiro *et al.* tried to combine mesenchymal migration with competing protrusions and chemotaxis by introducing the reaction kinetics model as a signal pathway.[117] However, in most of these approaches, the intricate and complex biochemical interactions between the attractants and the receptors to drive cell migrations are neglected.

The last category is about collective cell migration, in which cell-cell interaction and adhesion become important, and their simple migration behavior can also induce a complex migrating pattern of the collective cells. Cell alignments on curved surfaces or micro-patterned substrates are modelled by using the linear elasticity model and the generalized Potts model.[118-120] Albert and Schwarz studied both single-cell and collective-cell dynamics on adhesive micropatterns with the generalized Potts model (**Fig. 6C**).[119] The phase field model was used by Palmieri *et al*, to study the sensitive dependence of cell stiffness in collective cell migration, explaining the ballistic motion of cancer cells surrounded by normal cells (**Fig. 6D**).[114] Although the phase field model shows that the polarization field can be used to describe directional actin polymerization, it does not include the reaction kinetics of chemicals participating in polymerization and depolymerization, and their transport phenomena inside and outside the cells. Given the fact that cell migration is highly sensitive to the surrounding cells, development of models for collective cell migration will become more important for studying cell migration in tumor environments.

The biggest challenge in theoretical and numerical analysis for microengineered platforms is how we can identify essential parameters and build the minimal model from such complex, multi-dimensional microenvironment. Furthermore, intensive and extensive image analysis for cell tracking in experiments should be accompanied to aid development of theoretical and numerical studies. Since cell migration is observed in a limited field of view for a long time, individual cell identification and pattern recognition of cell trajectories become mandatory.[121] However, manual cell tracking bearing human errors and image restoration is still a popular method today. Recent studies show that machine learning can be used to identify cell morphologies in collective cell migrations, and also the cell migration patterns.[122-125] It is expected that machine learning can help recognize an individual movement pattern, distinguish between diffusive and superdiffusive motion, and predict a realistic migration in various microenvironments.

Conclusions

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Table 1. Summary of the recently developed microengineered platforms for cell migration in tumor environments

Types of tumor environment	*Microengineered cues [Ref]*	*Advantages of the microengineered platform*	*Perspective for future microengineered platform*
Physical microenvironment	Flow [6, 35]	. Microfluidic control of flow (***interstitial or bulk shear flow***) to study its effects on tumor cell migration and metastasis. . Physiologically relevant ***3D cell culture*** system	**Substrate materials:** · Verification of unknown mechanical and/or biochemical effects (confinement, stiffness, roughness, adhesion, etc.) of substrate materials (glass, polymer, hydrogel, paper, etc.) **Control of external stimulation:** · Maintaining stable gradient for long term experiments · Minimization of unwanted shear stress in the migration tracks · Spatial and temporal control of multiple cues and quantification of synergistic or competitive effects **Imaging and theoretical modeling:** · Tracking and understanding of dynamics of cellular migration under more *in-vivo* like, complex, multi-cue microenvironment **Applications** · Suggestion of new biological discovery beyond mere validation of already known soluble factors in tumor microenvironment · Translation towards clinical application based on cell sorting by motility differences
	Geometrical confinement [48,50]	. Control of degrees of ***confinement*** in dimensions close to single cell level. . Control of ***cell adhesion*** on substrates . Novel cell migration mechanisms such as mesenchymal-amoeboid transition and osmotic engine model	
	Topology [52]	. Control of the nanostructure density to form ***gradient of topography*** on substrates . New type of directed migration affected by topologically complex extracellular matrix (ECM)	
Multi-cue microenvironment	Haptotactic + chemotactic gradient [60]	. Microfluidic chamber for quantitative analysis of cell migration ***in combined response to substrate-bound and soluble chemokine gradients***	
	Confinement + obstacles + chemotactic gradient [61,62]	. ***Simplified design of the migration tracks*** to distinguish effects from obstacles such as cell-to-cell collision and narrow constrictions (~2 μm) from ***multiple migratory cues*** . New findings such as cell recognition and repair of nuclear envelopes during the migration	
	Multiple soluble cues + gradient of oxygen concentration [63]	. ***Easy assembly of paper stacks*** for imposing multiple cues and ***isolation of migrated cells***	
Intercellular microenvironment	Cross talk between two types of cells [71,75,80,89,90]	. ***Quantitative analysis of interaction between two different cell types*** . Integration of ***diverse migration tracks (1D, 2D, or 3D) between different cell types***	
	Cross talk among multiple types of cells [76,81,91]	. ***Spatial and temporal control of multi-cellular crosstalk*** by using microchannels, pillars and valves . ***Easy quantification of synergistic or competitive effects*** of multi-cellular interaction	
	Mechanical + biochemical stimulation [77]	. ***Simultaneous stimulation of mechanical (fluid shear stress and cyclic stretch) and biochemical microenvironment*** in a co-culture chip.	

This review has focused on the current state of knowledge regarding methods to investigate cell migration *in vitro* in tumor environments, with special attention paid to multi-cue microenvironments, including physical cues and interactions with multiple cell types. In order to optimize the quantitative usefulness of such studies, multiple cues should be controlled with high precision, and the analysis should differentiate quantitatively between combined or synergetic effects of several types of cues. Recent developments in microdevice fabrication and measurement technology have enabled researchers in this field to create not only sophisticated microenvironments for cell migration, but also analytical platforms for migration phenotype-based single cell sorting and analysis. There are still some drawbacks in the current microengineered system with room for improvements, such as interruptions of cell-to-cell signals by flow or adhesion of certain proteins to the surfaces of the device, and unwanted shear stress that may affect the cell migration ability. In addition, the actual gradient of chemicals across the cells can significantly differ from the predicted gradient in a smooth channel without cells for chemotaxis. Although we introduced examples of breakthrough that reveal novel mechanisms of cell migration in this review, many of the current microengineered system only proves to validate previously known cell-signaling factors relevant to cell migration, and lacks applications towards new therapeutic development. We summarized the advantages and perspectives of current system introduced in this review in Table 1. In the future, the authors anticipate that advance in clinical interpretation based on the cell migration ability, and theory and numerical analysis of single and collective cell migration will become increasingly important, and the field will cease to advance based solely on experiment.

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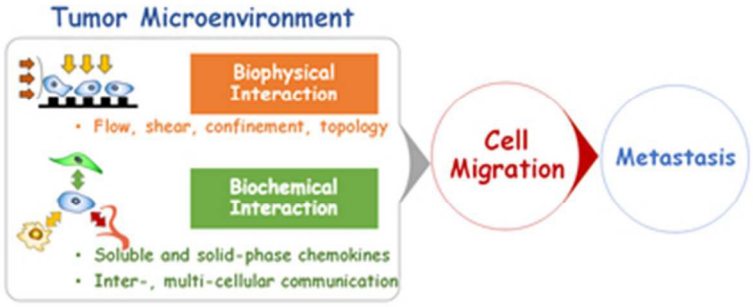
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